

2.5 Adding and Subtracting Polynomials

Lesson Introduction

 Benchmark	MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.		 Learning Target	I can add and subtract polynomials.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.		MA.912.AR.1.5 Divide polynomial expressions using long division, synthetic division, or algebraic manipulation. MA.912.AR.1.9 Apply previous understanding of rational number operations to add, subtract, multiply, and divide rational algebraic expressions.	
 Essential Questions	- How do you determine if two monomials are like terms? - How do you add and subtract like terms? - What is the same and different about adding and subtracting polynomials?			
 Lesson Narrative	In this lesson, students connect their prior knowledge of combining like terms to generate equivalent algebraic expressions from middle school to adding and subtracting polynomials. The 2.5.2 Guided Instruction and 2.5.3 Collaboration then introduce two strategies students can use when adding and subtracting polynomials before practicing on their own. The lesson then extends to connect the distribution property in Lesson 4 by distributing a negative to rewrite subtraction as addition of the opposite.			
 Component Summary	2.5.1 Warm-Up (~5 min.)	Analyzing a real-world math fail (<i>SE p. 80</i>)	2.5.4 Guided Practice (10 min.)	Structured practice adding and subtracting polynomials (<i>SE p. 83</i>)
	2.5.2 Guided Instruction (15-20 min.)	Introduction to adding polynomials (<i>SE p. 80</i>)	2.5.5 Your Turn! (10-15 min.)	Additional practice adding and subtracting polynomials (<i>SE p. 84</i>)
	2.5.3 Collaboration (15 min.)	Subtracting polynomials collaborative activity (<i>SE p. 82</i>)	2.5.6 Wrap-Up (~5 min.)	Reflecting on understanding and interest of adding and subtracting polynomials (<i>SE p. 85</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Like Terms <u>Additional Glossary:</u> - Polynomials		(Optional) Calculator (Optional) Highlighters	Consider making matching cards for 2.5.2 Guided Instruction question 2. Students often benefit from the kinesthetic movement of pairing cards.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may change the exponents when adding and subtracting like terms. - Students may not change subtracting a negative term to addition when subtracting polynomials.	- It is important that students have exposure to operating with all rational numbers from the beginning of the school year. To build fluency, teachers should include all types of rational numbers in these problems. - Continue highlighting the features of a polynomial throughout the lesson to solidify that knowledge from the previous lesson. - There are multiple opportunities for students to collaborate in this lesson, though it may be tempting for the teacher to just “tell” the students. Consider strategically partnering students so they can make these connections on their own and allow them to build a deeper understanding of the concepts.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 2.5.2 Guided Instruction and 2.5.3 Collaboration position students to compare different mathematical approaches to adding and subtracting polynomials. Apply this routine to position students to analyze different mathematical methods and how they connect to their current knowledge of mathematical operations and like terms.	MA.K12.MTR.4.1: Discussion and Reflection Students will be analyzing different strategies to add and subtract polynomials in the 2.5.2 Guided Instruction and the 2.5.3 Collaboration. Clarifications within this standard include analyzing the thinking of others and comparing the efficiency of methods used by others to solve problems.
 Differentiate Instruction	This lesson includes specific scaffolds and strategies based on common misconceptions students have when performing these operations. Using colors is helpful when identifying like terms and modeling their thinking. This is included throughout the lesson and can be continued by the teacher in class throughout student practice. If possible, it will be helpful if the teacher can provide highlighters or markers for the students to use. In lieu of those, symbols work (i.e., underlining, circling, or putting a triangle around sets of like terms).	
Lesson Notes:		

2.5.1 Warm-Up: Math Fail

Component Overview: Students will examine a real-world math fail and write a summary of what they notice.



2.5.1 Warm-Up: Math Fail

1. The classified advertisement shown is an example of a *math fail*.



What do you notice about the advertisement that makes it a *math fail*?

Answers will vary. It says you must be 18, not "at least" 18. If you have to be 18 and you must have 20 years experience, that would be impossible because you would not have even lived long enough to meet that requirement.



Consider using this routine while students analyze the classified advertisement. Position students to not only notice why it is a math fail but also wonder how the possible error was made when creating the advertisement for the waitress being hired.

2.5.2 Guided Instruction: Adding Polynomials

Component Overview: Students will be collaborating with a partner to identify like terms and how to add polynomials using like terms. Utilize the structure of this component to lead students to reach their own understanding by analyzing different methods to add polynomials. Use questioning if needed when pairs of students struggle to find an entry-point.



2.5.2 Guided Instruction: Adding Polynomials



Like terms are those whose variable(s) and their corresponding exponents are the same.

The following are examples of like terms.

Example 1	Example 2
$3.7mn^6$ and $-12mn^6$	$-cb^2a^{10}$, $81a^{10}cb^2$, and $\frac{1}{2}a^{10}b^2c$

- Discuss with your partner why $5ev^9$ and $5ev^3$ are not like terms. Summarize your discussion.

Answers will vary. The exponent on the variable 'v' is different.

- Draw an arrow to match the like terms between column A to column B. Then, check your matches with your partner.

Column A	Column B
$4xy^6$	$9yx$
$3xy$	x^6y
$\frac{11}{4}x^5y^5$	$0.2x^3y^2$
$4x^6y$	$-2.9x^2y^3$
$-1.6x^2y^3$	$-7y^6x$
$-6\frac{7}{8}x^3y^2$	$-3x^5y^5$



Provide students the opportunity to independently read, process, and answer the prompt in question 1 before discussing with a partner. Think-Pair-Share gives students three opportunities to make sense of information, first for independent learners, then pairing for verbal processing in order to give opportunities for more confident, precise responses when students share out.



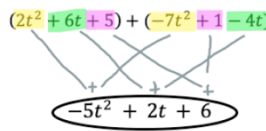
Discussion provides an auditory entry-point. Consider utilizing sentence stems and provide ample time for students to grapple with academic language:

- I think the two that match are _____ and _____ because ...
- I believe two monomials are like terms because ...
- I agree with you because ...
- I disagree with you because ...

2.5.2 Guided Instruction: Adding Polynomials (continued)

When adding and subtracting polynomials, only like terms can be combined.

3. Ziare and Colby used different strategies to add the polynomials $(2t^2 + 6t + 5)$ and $(-7t^2 + 1 - 4t)$.
- a. Work with your partner to describe what each student did and identify one helpful aspect used in their strategy.

$(2t^2 + 6t + 5) + (-7t^2 + 1 - 4t)$		
	Ziare's Strategy	Colby's Strategy
		$ \begin{array}{r} 2t^2 + 6t + 5 \\ + (-7t^2 - 4t + 1) \\ \hline -5t^2 + 2t + 6 \end{array} $
Description of the Strategy	Answers will vary. Ziare highlighted each like term in the same color and then added them one at a time to get the final answer.	Answers will vary. Colby stacked the like terms on top of each other and added to get the final answer.
What was helpful?	Answers will vary. It was helpful to make sure no terms were missed.	Answers will vary. It was helpful to make sure the signs weren't missed.



Utilize the structure of 2.5.2 Guided Instruction to foster students' meta-awareness while they compare different mathematical approaches to add polynomials.



After students complete the table, consider asking the following.

- Which strategy do you think you will use and why?
- Can you see why someone would use the other strategy to add polynomials?
- What about each strategy do you think is not helpful?
- In the example, colors were used to indicate like terms, how else could you symbolize like terms when you are working to make adding polynomials easier?

2.5.2 Guided Instruction: Adding Polynomials (continued)

- b. Find the sum of $(19h^4 - 2h + 25k) + (8h - 14k + h^4)$ using both Ziare's and Colby's strategies.

$(19h^4 - 2h + 25k) + (8h - 14k + h^4)$	
Ziare's Strategy	$(19h^4 - 2h + 25k) + (h^4 + 8h - 14k)$ $(19h^4) + (h^4) = 20h^4$ $(-2h) + (8h) = 6h$ $(25k) + (-14k) = 11k$ $20h^4 + 6h + 11k$
Colby's Strategy	$\begin{array}{r} 19h^4 - 2h + 25k \\ + \quad h^4 + 8h - 14k \\ \hline 20h^4 + 6h + 11k \end{array}$



Students are challenged to use both strategies to add two polynomials in the second part of question 3. By positioning students to use both methods, students must analyze and think deeply behind why both methods work mathematically.

4. Find the sum using your preferred method. Write the sum in standard form.
 $(-3p^3k - 2k^2 + 1.4p) + (2.7p^3k - 4.6p + 5.8k^2)$
 $-0.3p^3k + 3.8k^2 - 3.2p$

2.5.3 Collaboration: Subtracting Polynomials

Component Overview: Students will collaborate on the process of subtracting polynomials. Monitor students as they work in pairs to apply the knowledge gained from 2.5.2 Guided Instruction and apply it to subtracting polynomials. Use questioning as needed to position students to find an entry-point.



2.5.3 Collaboration: Subtracting Polynomials

The process for subtracting polynomials is similar to the one for adding polynomials.

- With your partner, find the difference of $(8x^5 + 4x^3 - 7)$ and $(11 - x^3 + 5x^5)$ by completing Ziare's and Colby's strategies.

$(8x^5 + 4x^3 - 7) - (11 - x^3 + 5x^5)$	
Ziare's Strategy	Colby's Strategy
$(8x^5) - (5x^5) = 3x^5$ $(4x^3) - (-x^3) = 5x^3$ $(-7) - (11) = -18$ $3x^5 + 5x^3 - 18$	$\begin{array}{r} 8x^5 \quad +4x^3 \quad -7 \\ -(5x^5) -(-x^3) -(+11) \\ \hline 3x^5 \quad +5x^3 \quad -18 \end{array}$



Utilize the structure of this component of the lesson to position students to find entry-points into subtracting polynomials. Encourage students to use the method that resonates with them the most of the three.



Students analyze different students' strategies for subtracting polynomials. To guide this reflection on mathematical thinking, consider asking the following:

- How do the strategies compare?
- Which method is more efficient and why?
- Do you think that one method is better than the other and if so, why?
- Which method do you plan on using and why?

2.5.3 Collaboration: Subtracting Polynomials (continued)

2. Oliver noticed another strategy that can work for subtraction.

He said, "I know the subtraction sign, or negative, can also mean 'opposite,' so I read the problem as $(8x^5 + 4x^3 - 7)$ plus the opposite of $(11 - x^3 + 5x^5)$."

He rewrote the problem to model his thinking.

$$(8x^5 + 4x^3 - 7) + (-11 + x^3 - 5x^5)$$

- a. Explain why you agree or disagree with Oliver's thinking.

Answers will vary. I agree with Oliver because the concept of subtraction is the same as adding a negative or opposite, therefore by changing the problem to addition and rewriting the second polynomial as its opposite, it would result in the same answer. I agree with Oliver because I checked his thinking with the problem and arrived at the same answer.

- b. Why might it be helpful to rewrite subtraction of polynomials as addition of polynomials?

Answers will vary. I think it would be helpful because it helps avoid any calculation mistakes when subtracting negatives. I think it would be helpful because addition is easier for me and I can stick with what I'm most comfortable with.

3. Find the difference of $(5m^2 - 4)$ and $(8m + 4 + 3m^2)$. Write the resulting polynomial in standard form.

$$2m^2 - 8m - 8$$



After students find the difference in question 3, consider asking the following:

- Which method did you use to solve and why?
- What makes subtraction seem "trickier" than addition?
- What error do you think you will have to be aware of the most while using your selected method?



Challenge students to think deeply about the mathematical logic behind all 3 methods presented by having them answer question 3 using all 3 methods and then comparing their answers.

2.5.4 Guided Practice: Adding and Subtracting Polynomials

Component Overview: For this component, students are applying what they learned to find the sum and difference of two polynomials. Introduce each question and have students determine the answer independently in an “I Do-We Do” approach and review the answer for each question.



2.5.4 Guided Practice: Adding and Subtracting Polynomials

Find each sum or difference. Write the result in standard form.

$$(14m^2n^2 + 2mn - 9) - (5m^2n^2 - 6mn + 17)$$

$$9m^2n^2 + 8mn - 26$$

$$(-2.3r^2 + 8.12pr - 0.4p) + (-7pr - 5.2r^2)$$

$$-7.5r^2 + 1.12pr - 0.4p$$



To ensure shared understanding, give students time to independently solve each question and then have a whole-group discussion about the answer and methods used. During independent work time, monitor and take note of students to probe during the whole-group discussion to promote the use of various methods.



Consider using different methods of presentation to best meet the students individually. Consider providing 3 options based on observations from previous lessons.

- Partner work
- Individual assignment
- Whole-group instruction

2.5.5 Your Turn!

Component Overview: Monitor students as they work independently to add and subtract polynomials. Use questioning to help students find entry-points when needed.



2.5.5 Your Turn!

Find each sum or difference. Write the result in standard form.

$$1. (ab^2 + 13b - 4a) - \left(7b + a + \frac{1}{3}ab^2\right)$$

$$\frac{2}{3}ab^2 - 5a + 6b$$

$$2. (8m^5 + 4m^3 - 9) - (11 + 5m^5 - m^3)$$

$$3m^5 + 5m^3 - 20$$

$$3. (-9d + 5d^2 + 3) + (5d - d^2 + 13)$$

$$4d^2 - 4d + 16$$

$$4. 3x^5y^3 + (x^3y^5 - 3x^5y^3)$$

$$x^3y^5$$

$$5. (7.1ct - 5) + (3t + 4ct^2)$$

$$4ct^2 + 7.1ct + 3t - 5$$



Consider providing students with choice by positioning them to choose 2 addition and 2 subtraction problems to complete to demonstrate their understanding of adding and subtracting polynomials.



If students finish early, challenge students to create an Error Analysis question using an error they saw or experienced during the day's lesson about adding and subtracting polynomials.

2.5.6 Wrap-Up: Reflection

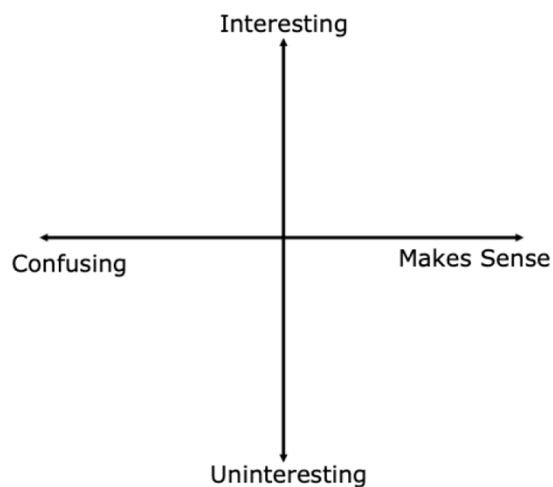
Component Overview: Students will assess their own understanding and provide feedback regarding the concepts from the lesson. Take note of the what the students find confusing to address those during another lesson or review.



2.5.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants to indicate how you feel about today's learning target, "I can add and subtract polynomials."

Answers will vary.



2. Describe why you plotted your point in that location.

Answers will vary.



Consider extending this activity by challenging students to provide an example and non-example of like terms.